Pre-Calculus 11 - Review Package



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

Factoring and Quadratic Equations Part 1

1. Factor fully. Write DNF if it does not factor.

   a) $9x^2y - 12xy$
   b) $2x(y-3) + 3(y-3)$
   c) $15a^3b^2 - 20a^2b^3$
   d) $4m(n+1) - 5(n+1)$
   e) $x^2 + 7x + 6$
   f) $x^2 - 8x + 15$
   g) $y^2 + 3y - 28$
   h) $m^2 - m - 72$
   i) $4x^2 + 9x - 12x - 27$
   j) $2x^2 - 10x - 28$
   k) $3x^2 - 10x + 8$
   l) $5x^2 + 13x - 6$
   m) $-20x^2 + 42x + 20$
   n) $100x^2 - 81y^2$
   o) $49a^2 - 16b^2$
   p) $2x^2 + 4x + 5$

2. Solve for x.

   a) $2x(x-7) = 0$
   b) $(4x-3)(x+2) = 0$
   c) $(6-5x)(8+3x) = 0$
   d) $5x(3x+1) = 0$
   e) $(x-9)(x+9) = 0$
   f) $(2x-5)(2x+5) = 0$
   g) $x^2 - 11x + 24 = 0$
   h) $x^2 + 6x - 16 = 0$
   i) $2x^2 - 5x - 3 = 0$
   j) $3x^2 + 10x - 8 = 0$
   k) $4x^2 - 25 = 0$
   l) $x^2 + 14x + 49 = 0$

3. Solve for x. Show at least one intermediate step. Round answers to the nearest tenths if necessary. If no real solutions exist, state no solutions.

   a) $x^2 + 10x - 24 = 0$
   b) $4x^2 + 3x - 10 = 0$
   c) $5x^2 + 2x - 1 = 0$
   d) $6x^2 + 3x + 1 = 0$
   e) $2x^2 - 7x - 4 = 0$
   f) $3x^2 - 4x - 2 = 0$

4. Solve for all values of x.

   a) $(2x+3)^2 = 3x^2 - 2$
   b) $4x^2 - 3x + 10 = 4x(x-6)$
   c) $(x-1)(x+4) = x^2 - 2x + 8$
   d) $2(x-3)^2 = x(x-2) + 12$
   e) $(3x-1)(x+2) = (x-2)(x+1)$
   f) $5x(x-1) = 3(x^2 + 2x - 4)$

5. For what values of b, will the expression be factorable?

   a) $3x^2 + bx + 10$
   b) $4x^2 + bx - 9$
   c) $2x^2 + bx - 5$

6. Solve for x.

   a) $(4x-5)(x+2) = (x-7)(x+1) + 17x$
   b) $3(x-2)(x+4) = 2x(x-1) + 10$

7. Find the value of a or k that makes the expression a perfect square trinomial.

   a) $ax^2 + 40x + 25$

   b) $9x^2 - 24x + k$

8. Fully factor the following.

   a) $3(7x-1)^2 + 4(7x-1) - 4$

   b) $4(a+b)^2 - 25(a-c)^2$

   c) $2(3x+2)^2 - 5(3x+2) - 12$

   d) $16(x-y)^2 - 9(x+2y)^2$

9. The height of a projectile can be modelled by $h(t) = -4.9t^2 + 10t + 3$, where h is height in meters and t is time in seconds. When does the projectile first hit the ground? Round answers to the nearest hundredths.

10. A rectangle has an area of 84 cm^2. The length of the rectangle is 5 cm more than its width. Set up a quadratic equation and find the dimensions of the rectangle.

11. The product of two consecutive positive odd integers is 143. If the smaller integer is n, write and solve an equation to find the two integers.

12. The hypotenuse of a right triangle is 1 cm longer than one leg. The other leg is 7 cm. Find the lengths of all three sides of the triangle.

Radical Expressions and Equations

13. State if the following are rational or irrational numbers. If rational, simplify.

a) $\sqrt[3]{0.027}$ b) $\sqrt{\frac{200}{5}}$ c) $\sqrt{144}$ d) $\sqrt{50}$

e) $\sqrt[3]{-64}$ f) $\sqrt{0.09}$ g) $\sqrt[4]{16}$ h) π

14. Write the following in ***radical form***.

a) $4^{\frac{2}{3}}$ b) $10^{\frac{1}{2}}$ c) $x^{\frac{3}{5}}$ d) $y^{\frac{5}{2}}$

e) $7^{\frac{3}{4}}$ f) $12^{\frac{1}{3}}$ g) $a^{\frac{2}{7}}$ h) $b^{\frac{4}{3}}$

15. Write the following in ***exponential form***.

a) $\sqrt[5]{x^3}$ b) $\sqrt[6]{y}$ c) $\sqrt{z^7}$ d) $\sqrt[3]{m^2}$

e) $\sqrt[4]{a^5}$ f) $\sqrt[7]{b^3}$ g) $\sqrt[8]{c}$ h) $\sqrt{10^3}$

16. Write the following mixed radicals as an entire radical.

a) $3x\sqrt{7x^2y}$ b) $2y^2\sqrt[3]{5x^2y}$ c) $4\sqrt{5}$ d) $2a\sqrt{3a}$

e) $5m\sqrt[3]{2m}$ f) $x^2\sqrt{x}$ g) $-3\sqrt[3]{4}$ h) $6y\sqrt{2x}$

17. Simplify the following radical expressions.

a) $\sqrt{50x^2y^3}$ b) $2\sqrt[3]{81}$ c) $\sqrt{48a^4b}$ d) $\sqrt[3]{-32x^5}$

e) $3\sqrt{20}$ f) $\sqrt{72m^5n^6}$ g) $-4\sqrt[3]{250}$ h) $\sqrt{108x^3y^2}$

18. Simplify the following radical expressions.

a) $322\sqrt{6} - 23\sqrt{7} + 345\sqrt{6} + 17\sqrt{7}$ b) $-3\sqrt[3]{x} + 4\sqrt[4]{x} - 5\sqrt[4]{x}$

c) $5\sqrt{2} - 3\sqrt{3} + 4\sqrt{2} + 8\sqrt{3}$ d) $6\sqrt{x} - 2\sqrt{y} - 9\sqrt{x} + 4\sqrt{y}$

e) $\sqrt{12} + \sqrt{27} - \sqrt{75}$ f) $2\sqrt{8} - 3\sqrt{18} + \sqrt{50}$

g) $\sqrt[3]{16} + 2\sqrt[3]{54}$ h) $4\sqrt{20} - 2\sqrt{45}$

19. Determine the restriction on the variables in order for the expression to exist.

a) $\sqrt[7]{x^3y^2}$ b) $\sqrt[6]{x-6}$ c) $\sqrt{2x+8}$

d) $\sqrt[4]{10-5x}$ e) $\sqrt[3]{x-1}$ f) $\sqrt{-3x}$

20. Simplify the following expressions.

a) $20\sqrt{32a} + 45\sqrt{18a} - 5\sqrt{50a}$

b) $4\sqrt[3]{128} - 3\sqrt[3]{54} + 2\sqrt[3]{8}$

c) $3\sqrt{75x^3} - x\sqrt{12x} + 2\sqrt{27x^3}$

d) $2\sqrt[4]{48} + \sqrt[4]{243} - 3\sqrt[4]{3}$

e) $\sqrt{20} + \sqrt{45} - \sqrt{80}$

f) $x\sqrt{18x} - 3\sqrt{8x^3} + \sqrt{50x^3}$

21. Multiply and simplify the following expressions.

a) $\sqrt{3x}(2\sqrt{x} + \sqrt{12x})$

b) $(x\sqrt{5} + 2\sqrt{x})(\sqrt{x} + 3\sqrt{5})$

c) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$

d) $(2\sqrt{a} - 3)^2$

e) $\sqrt{6}(3\sqrt{2} - 2\sqrt{3})$

f) $(\sqrt{x} + \sqrt{y})(\sqrt{x} - \sqrt{y})$

22. Simplify the following by rationalizing the denominator.

a) $\dfrac{16x^3\sqrt{50y}}{4x\sqrt{2}}$

b) $\dfrac{5\sqrt{7} + 4}{\sqrt{7}}$

c) $\dfrac{10}{\sqrt{5}}$

d) $\dfrac{3\sqrt{2}}{2\sqrt{3}}$

e) $\dfrac{\sqrt{8x^3}}{\sqrt{2x}}$

f) $\dfrac{4\sqrt{5} - \sqrt{2}}{\sqrt{10}}$

23. Solve for all possible values of x in the following. Leave answers as exact values.

a) $\sqrt[3]{x - 2} = -3$

b) $\sqrt{x + 2} = 2x - 3$

c) $\sqrt{3x + 1} = 4$

d) $\sqrt{5x - 4} = x$

e) $2\sqrt{x - 1} = 6$

f) $\sqrt{x^2 + 5} = x + 1$

24. Rationalize the denominator and simplify the following.

a) $\dfrac{3\sqrt{x} + 2y}{\sqrt{y} + 4\sqrt{x}}$

b) $\dfrac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

25. Simplify using the exponent laws. Leave your answer with positive exponents only.

a) $\dfrac{\sqrt{6}x^{\frac{5}{3}}y^{-2} \cdot \sqrt{3}x^{-\frac{1}{6}}y^3}{9x^{\frac{1}{2}}y^{-1}}$

b) $\left(\dfrac{8x^3y^{-6}}{27x^{-3}y^3}\right)^{-\frac{1}{3}}$

26. Solve for all possible values of x in the following. Leave answers as exact values.

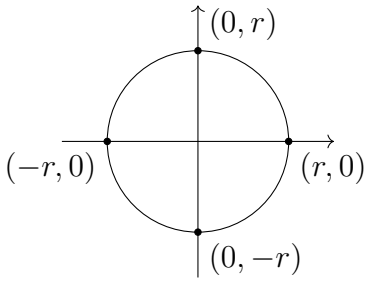
a) $\sqrt{5x + 6} - \sqrt{3x - 2} = 2$

b) $\sqrt{2x + 3} + \sqrt{x - 2} = 4$

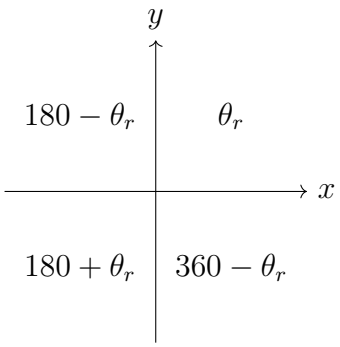
Trigonometry

Reference Material

	30°	45°	60°
$\sin\theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
$\tan\theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

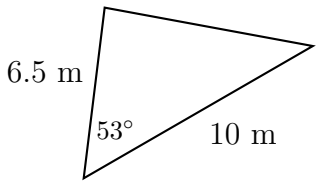


- $\sin\theta = \dfrac{y}{r}$
- $\cos\theta = \dfrac{x}{r}$
- $\tan\theta = \dfrac{y}{x}$

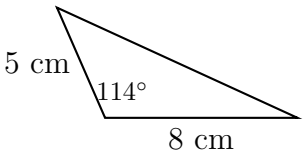


27. Find the area of the following triangles. Round final answers to the hundredths.

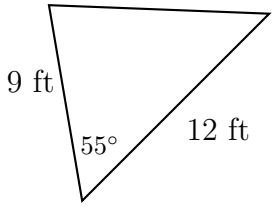
a)



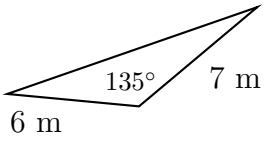
b)



c)

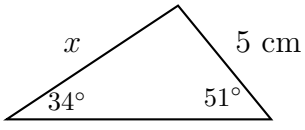


d)

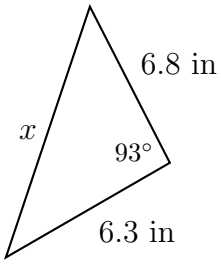


28. Find the indicated side x or angle θ. Round answers to the nearest tenths.

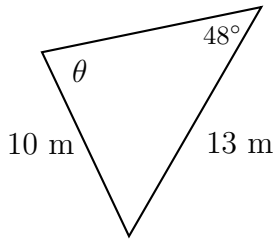
a)



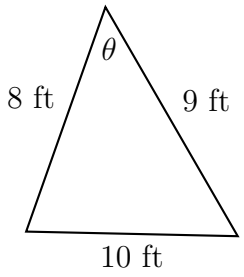
b)



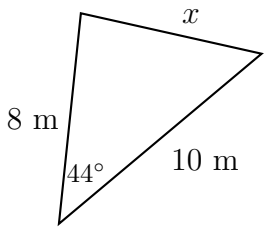
c)



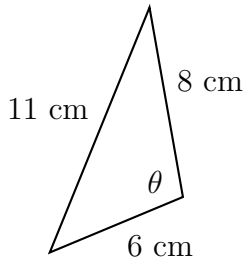
d)



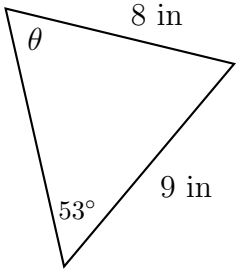
e)



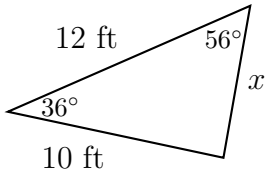
f)



g)



h)



29. Sketch the following angles in standard position.

a) 120° b) −330° c) 225° d) −45°



e) 315° f) −150° g) 60° h) 270°



30. Given the following θ in standard position, determine the reference angle.

a) 240° b) −315° c) 150° d) −60°
e) 330° f) 200° g) −120° h) 135°

31. Find one positive and one negative co-terminal angle.

a) 310° b) 45° c) −120° d) 225°
e) −50° f) 150° g) 330° h) −300°

32. Determine which quadrant(s) an angle θ in standard position would terminate in given the conditions.

    a) $\sin\theta > 0$ and $\tan\theta < 0$
    b) $\cos\theta > 0$ and $\tan\theta > 0$
    c) $\sin\theta < 0$ and $\tan\theta > 0$
    d) $\tan\theta < 0$ and $\cos\theta > 0$
    e) $\cos\theta > 0$ and $\sin\theta < 0$
    f) $\tan\theta > 0$ and $\sin\theta > 0$
    g) $\cos\theta < 0$ and $\tan\theta > 0$
    h) $\sin\theta < 0$ and $\cos\theta < 0$

33. Evaluate the following without a calculator.

    a) $\sin 225°$
    b) $\cos 300°$
    c) $\tan 180°$
    d) $\sin 150°$
    e) $\cos 135°$
    f) $\tan 330°$
    g) $\cos 270°$
    h) $\sin 315°$
    i) $\tan 240°$

34. Given that the point P is on the terminal arm of an angle, θ, in standard position. Determine the exact values of $\sin\theta, \cos\theta$, and $\tan\theta$.

    a) $P(5, -12)$
    b) $P(-3, 4)$
    c) $P(-8, -15)$

35. Find the exact values of the other two primary trigonometric ratios.

    a) $\cos\theta = \dfrac{-4}{\sqrt{22}}$, $\sin\theta < 0$
    b) $\sin\theta = \dfrac{3}{5}$, $\tan\theta < 0$
    c) $\tan\theta = \dfrac{12}{5}$, $\cos\theta < 0$

36. Solve the following for $0° \leq \theta < 360°$, without a calculator.

    a) $\tan\theta = \sqrt{3}$
    b) $\sin\theta = 0$
    c) $\cos\theta = -\dfrac{\sqrt{3}}{2}$
    d) $\tan\theta =$ undefined
    e) $\sin\theta = -\dfrac{1}{2}$
    f) $\cos\theta = \dfrac{1}{\sqrt{2}}$
    g) $\tan\theta = -1$
    h) $\cos\theta = -1$
    i) $\sin\theta = \dfrac{\sqrt{3}}{2}$

37. Solve the following.

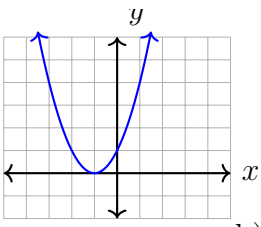
    a) Find **general solutions** to $\cos\theta = 0.123$. Round final answers to 2 decimal spots.
    b) Solve $\tan\theta = -1$, given $-180° \leq \theta < 180°$.
    c) Solve $(\cos\theta + 1)(2\sin\theta - 1) = 0$, given $0° \leq \theta < 360°$.

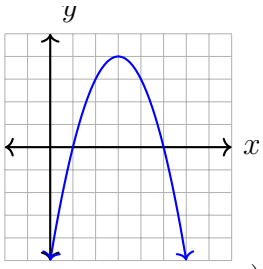
38. Two planes depart at the same time from the same location. One travels 800 km/h at a bearing of 060°, and the other travels 850 km/h at a bearing of 300°. How far apart are the two planes after 1 hour? Round to the nearest km.

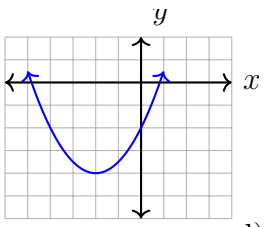
39. Given $\angle A = 40°, a = 30$ cm, and $b = 40$ cm. How many possible triangles are there? Explain briefly, no need to solve.

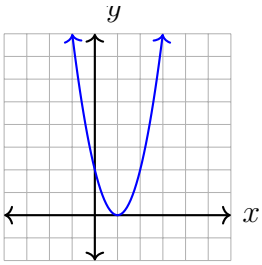
Quadratic Functions and Equations Part 2

40. Given the graph of the quadratic function below, identify the coordinates of the vertex, the equation of the axis of symmetry, and the domain and range.

a) 

b) 

c) 

d) 

41. State the vertex, axis of symmetry, and direction of opening for the following functions.

a) $y = -2(x-3)^2 + 4$
b) $y = \frac{1}{2}(x+1)^2 - 5$
c) $y = 3(x-5)^2 - 1$
d) $y = -(x+4)^2 + 7$
e) $y = 0.25(x-2)^2$
f) $y = -5x^2 + 3$
g) $y = \frac{2}{3}(x+6)^2 - 2$
h) $y = -1.5(x-1)^2 + 8$

42. Solve the following equations using the square root method. Leave answers in exact form.

a) $(x-4)^2 = 25$
b) $2(x+3)^2 - 10 = 0$
c) $3(x-1)^2 = 27$
d) $-4(x+2)^2 + 16 = 0$
e) $\frac{1}{2}(x-5)^2 - 8 = 0$
f) $(x+6)^2 + 5 = 20$
g) $5(x-2)^2 = 40$
h) $-2(x-3)^2 + 18 = 0$

43. Describe the transformations required to change the graph of $y = x^2$ into the graph of the given function.

a) $y = -3(x+2)^2 - 5$
b) $y = \frac{1}{4}(x-1)^2 + 3$
c) $y = -2(x+4)^2$

44. Determine the equation of the quadratic function in vertex form given the following properties.

a) Vertex $(-3, 0)$, y-intercept $(0, -9)$
b) Vertex $(2, 5)$, passes through $(4, -3)$
c) Vertex $(-1, -4)$, passes through $(1, 8)$

45. Convert the following to Vertex Form by **completing the square**.

a) $y = x^2 - 10x + 19$
b) $y = x^2 + 6x - 2$
c) $y = 2x^2 - 12x + 11$
d) $y = -3x^2 - 18x - 5$
e) $y = \frac{1}{2}x^2 + 4x + 3$
f) $y = -x^2 + 8x - 10$

46. Find the vertex of the following using any method.

a) $y = -2x^2 - 8x + 5$ b) $y = 3x^2 - 12x + 1$ c) $y = x^2 + 5x - 4$

47. Use the discriminant to determine the nature of the roots (do not solve).

a) $2x^2 - 3x + 5 = 0$ b) $x^2 - 6x + 9 = 0$ c) $3x^2 + 4x - 2 = 0$

48. Find the value(s) of k such that the equation has exactly **one real root**.

a) $x^2 + kx + 9 = 0$ b) $2x^2 - 8x + k = 0$ c) $kx^2 + 12x + 4 = 0$

49. Solve the following quadratic equations. Write answers in exact form.

a) $\frac{1}{3}(2x - 3)^2 = 8$ b) $-2(3x + 1)^2 + 14 = 0$

50. A concert venue has 3,000 seats and usually sells 2000 tickets when the ticket price is \$120. Market research indicates that for every \$10 increase in ticket price, 100 less tickets are sold. Determine the ticket price that will maximize revenue and how many tickets will be sold at that price.

51. A farmer has 250 meters of fencing to enclose a rectangular field. She also wants to add a fence dividing the field in half parallel to one side. Determine the dimensions that maximize the total area.

52. A stadium scoreboard has a parabolic shape with a width of 20 m and reaches a maximum height of 15 m at its center. Find the height of the scoreboard at a point 5 m from one of the sides.

53. A factory's daily production cost for widgets is modeled by $C(x) = 500 - 8x + 0.2x^2$, where x is the number of widgets produced. Determine the number of widgets that minimizes the cost.

54. A hotel with 400 rooms is typically 80% full when the nightly rate is \$150. Market research indicates that for every \$15 increase in the room rate, 10 fewer rooms are booked. Determine the room rate and number of rooms rented that maximize revenue.

55. A landscaper is designing a rectangular park using 120 meters of fencing. In addition to the perimeter, she plans to install two parallel interior fences dividing the park into three equal sections. Determine the dimensions that maximize the total area and calculate that area.

Rational Expressions and Equations

56. State the non-permissible values (NPVs) for the following rational expressions.

a) $\dfrac{5x}{4xy^3}$ b) $\dfrac{x-2}{x^2+7x+10}$ c) $\dfrac{3a}{a^2-9}$ d) $\dfrac{y+1}{y^2-4y}$

e) $\dfrac{2x}{x^2-16}$ f) $\dfrac{x+5}{x^2+5x+6}$ g) $\dfrac{7}{3x^2y}$ h) $\dfrac{x-1}{2x^2-8x}$

57. Simplify the following rational expressions. State the NPVs.

a) $\dfrac{18a^3b^2}{21a^2b^5}$ b) $\dfrac{3x-12}{x^2-16}$

c) $\dfrac{x^2+5x+6}{x^2+2x}$ d) $\dfrac{2x^2-8}{4x+8}$

e) $\dfrac{x^2-9x+20}{x^2-25}$ f) $\dfrac{5x^2y^4}{15xy^6}$

g) $\dfrac{x^2-x-12}{x^2-9}$ h) $\dfrac{4x-20}{x^2-10x+25}$

58. Perform the indicated operation and simplify. State the NPVs.

a) $\dfrac{x^2+2x}{x^2-9}\cdot\dfrac{3x-9}{4x}$ b) $\dfrac{x^2+5x+6}{x^2-4}\div\dfrac{x+3}{x-2}$

c) $\dfrac{5x}{x-3}+\dfrac{15}{x-3}$ d) $\dfrac{x^2+3x}{x+4}-\dfrac{2x+12}{x+4}$

e) $\dfrac{2x}{x+1}\cdot\dfrac{x^2-1}{4x^2}$ f) $\dfrac{x^2-25}{x^2+5x}\div\dfrac{x-5}{x}$

g) $\dfrac{4x}{x-2}-\dfrac{8}{x-2}$ h) $\dfrac{3x^2}{x+5}+\dfrac{15x}{x+5}$

59. Perform the indicated operations and simplify completely. State the NPVs.

a) $\dfrac{2x^2+5x-3}{x^2-9}\cdot\dfrac{12-4x}{2x^2-x}$ b) $\dfrac{y^2-y-12}{y^2-y-20}\div\dfrac{y^2+y-6}{y^2+2y-15}$

c) $\dfrac{4}{x^2-4}+\dfrac{2}{x^2+2x}$ d) $\dfrac{3}{x+4}-\dfrac{1}{x-2}$

e) $\dfrac{x^2-x-6}{x^2-4}\cdot\dfrac{x^2+4x+4}{x^2-9}$ f) $\dfrac{5}{x-3}-\dfrac{2}{x+1}$

60. Solve the following rational equations.

a) $\frac{2}{x} = \frac{x+3}{5}$

b) $\frac{x}{x-2} + 1 = \frac{6}{x-2}$

c) $\frac{3}{x+1} = \frac{2}{x-1}$

d) $\frac{x}{x-4} - \frac{4}{x-4} = 2$

e) $\frac{1}{x} + \frac{1}{2x} = 3$

f) $\frac{x+2}{x-1} = \frac{x+4}{x+2}$

61. Simplify the following mixed operations expression completely.

a) $\left(\frac{x}{x^2-25} - \frac{2}{x^2-5x}\right) \div \frac{x^2+2x}{x^2+5x}$

b) $\left(\frac{3}{x-2} + \frac{1}{x+2}\right) \cdot \frac{x^2-4}{4x+4}$

62. Solve the following rational equation algebraically.

a) $\frac{x}{x-3} - \frac{2}{x+1} = \frac{12}{x^2-2x-3}$

b) $\frac{2}{x^2-1} + \frac{1}{x-1} = \frac{1}{x+1}$

63. The quotient of a number and two is added to the quotient of six and the difference of the number and two. If the resulting sum is five, set up a rational equation and find all possible values for the number.

64. It takes Person A 4 hours to paint a room alone. Person B can paint the same room in 6 hours alone. Write a rational equation to find how long it takes them to paint the room working together, and solve it.

65. A river has a current of 4 km/h. A boat can travel 30 km downstream in the same amount of time it takes to travel 18 km upstream. Let x be the speed of the boat in still water. Write a rational equation to model this situation and solve for the speed of the boat.

Solutions

Factoring and Quadratic Equations Part 1

1. a) $3xy(3x-4)$ b) $(2x+3)(y-3)$ c) $5a^2b^2(3a-4b)$
   d) $(4m-5)(n+1)$ e) $(x+6)(x+1)$ f) $(x-5)(x-3)$
   g) $(y+7)(y-4)$ h) $(m-9)(m+8)$ i) $(4x+9)(x-3)$
   j) $2(x-7)(x+2)$ k) $(3x-4)(x-2)$ l) $(5x-2)(x+3)$
   m) $-2(5x+2)(2x-5)$ n) $(10x-9y)(10x+9y)$ o) $(7a-4b)(7a+4b)$
   p) DNF

2. a) $x=0,7$ b) $x=\frac{3}{4},-2$ c) $x=\frac{6}{5},-\frac{8}{3}$
   d) $x=0,-\frac{1}{3}$ e) $x=9,-9$ f) $x=\frac{5}{2},-\frac{5}{2}$
   g) $x=8,3$ h) $x=-8,2$ i) $x=3,-\frac{1}{2}$
   j) $x=\frac{2}{3},-4$ k) $x=\frac{5}{2},-\frac{5}{2}$ l) $x=-7$

3. a) $x=2,-12$ b) $x=1.3,-2.0$
   c) $x=0.3,-0.7$ d) No real solutions
   e) $x=4,-0.5$ f) $x=1.7,-0.4$

4. a) $x=-1,-11$ b) $x=-\frac{10}{21}$
   c) $x=6$ d) $x=2,8$
   e) $x=0,\frac{3}{2}$ f) $x=3,-2$

5. a) $b\in\{\pm11,\pm13,\pm17,\pm31\}$ b) $b\in\{0,\pm5,\pm9,\pm16,\pm35\}$
   c) $b\in\{\pm3,\pm9\}$

6. a) $x=-\frac{1}{3},3$ b) $x=-7,2$

7. a) $a=16$ b) $k=16$

8. a) $(21x-5)(7x+1)$ b) $(-3a+2b+5c)(7a+2b-5c)$
   c) $(3x-2)(6x+7)$ d) $(x-10y)(7x+2y)$

9. $t\approx 2.31$ seconds

10. Length = 12 cm, Width = 7 cm

11. The integers are 11 and 13.

12. The sides are 7 cm, 24 cm, 25 cm.

Radical Expressions and Equations

13. a) Rational: 0.3 b) Irrational: $2\sqrt{10}$ c) Rational: 12 d) Irrational: $5\sqrt{2}$
e) Rational: -4 f) Rational: 0.3 g) Rational: 2 h) Irrational

14. a) $(\sqrt[3]{4})^2$ b) $\sqrt{10}$ c) $\sqrt[5]{x^3}$ d) $\sqrt{y^5}$
e) $(\sqrt[4]{7})^3$ f) $\sqrt[3]{12}$ g) $\sqrt[7]{a^2}$ h) $\sqrt[3]{b^4}$

15. a) $x^{\frac{3}{5}}$ b) $y^{\frac{1}{6}}$ c) $z^{\frac{7}{2}}$ d) $m^{\frac{2}{3}}$
e) $a^{\frac{5}{4}}$ f) $b^{\frac{3}{7}}$ g) $c^{\frac{1}{8}}$ h) $10^{\frac{3}{2}}$

16. a) $\sqrt{63x^4y}$ b) $\sqrt[3]{40x^2y^7}$ c) $\sqrt{80}$ d) $\sqrt{12a^3}$
e) $\sqrt[3]{250m^4}$ f) $\sqrt{x^5}$ g) $\sqrt[3]{-108}$ h) $\sqrt{72xy^2}$

17. a) $5xy\sqrt{2y}$ b) $6\sqrt[3]{3}$ c) $4a^2\sqrt{3b}$ d) $-2x\sqrt[3]{4x^2}$
e) $6\sqrt{5}$ f) $6m^2n^3\sqrt{2m}$ g) $-20\sqrt[3]{2}$ h) $6xy\sqrt{3x}$

18. a) $667\sqrt{6} - 6\sqrt{7}$ b) $-3\sqrt[3]{x} - \sqrt[4]{x}$
c) $9\sqrt{2} + 5\sqrt{3}$ d) $-3\sqrt{x} + 2\sqrt{y}$
e) 0 f) 0
g) $8\sqrt[3]{2}$ h) $2\sqrt{5}$

19. a) $x \in \mathbb{R}, y \in \mathbb{R}$ b) $x \geq 6$ c) $x \geq -4$
d) $x \leq 2$ e) $x \in \mathbb{R}$ f) $x \leq 0$

20. a) $190\sqrt{2a}$ b) $7\sqrt[3]{2} + 4$
c) $13x\sqrt{3x}$ d) $8\sqrt[4]{3}$
e) $\sqrt{5}$ f) $2x\sqrt{2x}$

21. a) $2x\sqrt{3} + 6x$ b) $(x+6)\sqrt{5x} + 17x$
c) -1 d) $4a - 12\sqrt{a} + 9$
e) $6\sqrt{3} - 6\sqrt{2}$ f) $x - y$

22. a) $20x^2\sqrt{y}$ b) $5 + \frac{4\sqrt{7}}{7}$ c) $2\sqrt{5}$
d) $\frac{\sqrt{6}}{2}$ e) $2x$ f) $2\sqrt{2} - \frac{\sqrt{5}}{5}$

23. a) $x = -25$ b) $x = \frac{13+\sqrt{57}}{8}$
c) $x = 5$ d) $x = 1, 4$
e) $x = 10$ f) $x = 2$

24. a) $\dfrac{3\sqrt{xy} - 12x + 2y\sqrt{y} - 8y\sqrt{x}}{y - 16x}$ b) $5 - 2\sqrt{6}$

25. a) $\dfrac{\sqrt{2}xy^2}{3}$ b) $\dfrac{3y^3}{2x^2}$

26. a) $x = 6, 2$ b) $x = 3$

Trigonometry

27. a) 25.96 m^2 b) 18.27 cm^2 c) 44.23 ft^2 d) 14.85 m^2

28. a) $x \approx 6.9$ cm b) $x \approx 9.5$ in c) $\theta \approx 75.0°$ d) $\theta \approx 71.8°$
e) $x \approx 7.0$ m f) $\theta \approx 102.6°$ g) $\theta \approx 63.9°$ h) $x \approx 7.1$ ft

29. (Graphical answers) (a) Q2 (b) Q1 (c) Q3 (d) Q4 (e) Q4 (f) Q3 (g) Q1 (h) negative y-axis

30. a) 60° b) 45° c) 30° d) 60°
e) 30° f) 20° g) 60° h) 45°

31. (Answers vary, examples shown)
a) $670°, -50°$ b) $405°, -315°$ c) $240°, -480°$ d) $585°, -135°$
e) $310°, -410°$ f) $510°, -210°$ g) $690°, -30°$ h) $60°, -660°$

32. a) Quadrant II b) Quadrant I
c) Quadrant III d) Quadrant II
e) Quadrant IV f) Quadrant IV
g) Quadrant III h) Quadrant III

33. a) $-\frac{1}{\sqrt{2}}$ b) $\frac{1}{2}$ c) 0 d) $\frac{1}{2}$
e) $-\frac{1}{\sqrt{2}}$ f) $-\frac{1}{\sqrt{3}}$ g) 0 h) $-\frac{1}{\sqrt{2}}$
i) $\sqrt{3}$

34. a) $\sin\theta = -\frac{12}{13}, \cos\theta = \frac{5}{13}, \tan\theta = -\frac{12}{5}$ b) $\sin\theta = \frac{4}{5}, \cos\theta = -\frac{3}{5}, \tan\theta = -\frac{4}{3}$
c) $\sin\theta = -\frac{15}{17}, \cos\theta = -\frac{8}{17}, \tan\theta = \frac{15}{8}$

35. a) $\sin\theta = -\frac{\sqrt{6}}{\sqrt{22}}, \tan\theta = \frac{\sqrt{6}}{4}$ b) $\cos\theta = -\frac{4}{5}, \tan\theta = -\frac{3}{4}$
c) $\sin\theta = -\frac{12}{13}, \cos\theta = -\frac{5}{13}$

36. a) $60°, 240°$ b) $0°, 180°$ c) $150°, 210°$
d) $90°, 270°$ e) $210°, 330°$ f) $45°, 315°$
g) $135°, 315°$ h) $180°$ i) $60°, 120°$

37. a) $\theta \approx 82.93° + 360°n$ and $277.07° + 360°n$, where $n \in \mathbb{I}$
b) $\theta = -45°, 135°$
c) $\theta = 30°, 150°, 180°$

38. 1429 km

39. **2 possible triangles.** The height of the triangle is $40 \sin 40° \approx 25.7$ cm. Because $h < a < b$ $(25.7 < 30 < 40)$, the ambiguous case applies.

Quadratic Functions and Equations Part 2

40. a) V: $(-1, 0)$, AoS: $x = -1$, D: $x \in \mathbb{R}$, R: $y \geq 0$ b) V: $(3, 4)$, AoS: $x = 3$, D: $x \in \mathbb{R}$, R: $y \leq 4$ c) V: $(-2, -4)$, AoS: $x = -2$, D: $x \in \mathbb{R}$, R: $y \geq -4$ d) V: $(1, 2)$, AoS: $x = 1$, D: $x \in \mathbb{R}$, R: $y \leq 2$

41. a) V: $(3, 4)$, AoS: $x = 3$, Down
b) V: $(-1, -5)$, AoS: $x = -1$, Up
c) V: $(5, -1)$, AoS: $x = 5$, Up
d) V: $(-4, 7)$, AoS: $x = -4$, Down
e) V: $(2, 0)$, AoS: $x = 2$, Up
f) V: $(0, 3)$, AoS: $x = 0$, Down
g) V: $(-6, -2)$, AoS: $x = -6$, Up
h) V: $(1, 8)$, AoS: $x = 1$, Down

42. a) $x = 9, -1$ b) $x = -3 \pm \sqrt{5}$ c) $x = 4, -2$
d) $x = 0, -4$ e) $x = 9, 1$ f) $x = -6 \pm \sqrt{15}$
g) $x = 2 \pm 2\sqrt{2}$ h) $x = 6, 0$

43. a) Refl x-axis, V-stretch factor 3, L 2, D 5
b) V-compression factor 1/4, R 1, U 3
c) Refl x-axis, V-stretch factor 2, L 4

44. a) $y = -(x + 3)^2$ b) $y = -2(x - 2)^2 + 5$ c) $y = 3(x + 1)^2 - 4$

45. a) $y = (x - 5)^2 - 6$ b) $y = (x + 3)^2 - 11$ c) $y = 2(x - 3)^2 - 7$
d) $y = -3(x + 3)^2 + 22$ e) $y = \frac{1}{2}(x + 4)^2 - 5$ f) $y = -(x - 4)^2 + 6$

46. a) $(-2, 13)$ b) $(2, -11)$ c) $(-2.5, -10.25)$

47. a) No Real Roots b) 1 Real Root c) 2 Real Roots

48. a) $k = \pm 6$ b) $k = 8$ c) $k = 9$

49. a) $x = \frac{3 \pm 2\sqrt{6}}{2}$ b) $x = \frac{-1 \pm \sqrt{7}}{3}$

50. **Ticket Price:** \$160 **Tickets Sold:** 1600

51. 62.5 m by 41.67 m (or $\frac{125}{3}$ m)

52. 11.25 m.

53. number of widgets to minimize costs is 20. And the minimum cost would be 420.

54. room rate is $(150 + 15 \cdot 11) = \$315$ and the number of rooms is $(320 - 10 \cdot 11) = 210$.

55. The dimensions are 15 m by 30 m, and the maximum area is 450 m^2.

Rational Expressions and Equations

56. a) $x \neq 0, \quad y \neq 0$ b) $x \neq -5, -2$ c) $a \neq \pm 3$ d) $y \neq 0, 4$
e) $x \neq \pm 4$ f) $x \neq -3, -2$ g) $x \neq 0, y \neq 0$ h) $x \neq 0, 4$

57. a) $\dfrac{2a}{3b^3}, \quad a \neq 0, b \neq 0$ b) $\dfrac{3}{x+4}, \quad x \neq \pm 4$
c) $\dfrac{x+3}{x}, \quad x \neq 0, -2$ d) $\dfrac{x-2}{2}, \quad x \neq -2$
e) $\dfrac{x-4}{x+5}, \quad x \neq \pm 5$ f) $\dfrac{x}{3y^2}, \quad x \neq 0, y \neq 0$
g) $\dfrac{x-4}{x-3}, \quad x \neq \pm 3$ h) $\dfrac{4}{x-5}, \quad x \neq 5$

58. a) $\dfrac{3(x+2)}{4(x+3)}, \quad x \neq \pm 3, 0$ b) $1, \quad x \neq \pm 2, -3$
c) $\dfrac{5(x+3)}{x-3}, \quad x \neq 3$ d) $x-3, \quad x \neq -4$
e) $\dfrac{x-1}{2x}, \quad x \neq 0, -1$ f) $\dfrac{x+5}{x}, \quad x \neq 0, \pm 5$
g) $4, \quad x \neq 2$ h) $3x, \quad x \neq -5$

59. a) $-\dfrac{4}{x}, \quad x \neq \pm 3, 0, \frac{1}{2}$ b) $\dfrac{(y-4)(y+5)(y-3)}{(y-5)(y+4)(y-2)}, \quad y \neq 5, -4, -5, 3, -3, 2$
c) $\dfrac{6x-4}{x(x-2)(x+2)}, \quad x \neq \pm 2, 0$ d) $\dfrac{2x-10}{(x+4)(x-2)}, \quad x \neq -4, 2$
e) $\dfrac{x+2}{x-3}, \quad x \neq \pm 2, \pm 3$ f) $\dfrac{3x+11}{(x-3)(x+1)}, \quad x \neq -1, 3$

60. a) $x = -5, 2$ b) $x = 4$
c) $x = 5$ d) No solution (Extraneous $x = 4$)
e) $x = \frac{1}{2}$ f) $x = 8$

61. a) $\dfrac{x^2 - 2x - 10}{(x-5)(x)(x+2)}$ b) 1

62. a) $x = -2$ (Root $x = 3$ is extraneous) b) $x = -2$ (Root $x = 1$ is extraneous)

63. The number can be 4 or 8.

64. $t = 2.4$ hours

65. 16 km/h